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המכון למחקרי נפט וגיאופיסיקה
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

SOUTH NETANYA AREA

MAGNETO-TELLURIC SURVEY
(FIRST PHASE)

A. KL'ANG

Prepared for
SONOL ISRAEL LTD

August, 1974

MT/118/74

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

ISRAEL

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M.T.P. Nos. 1 - 4

ABSTRACT

Results of four magneto-telluric measurements conducted south of Netanya as a first phase of a larger scale magneto-telluric survey are presented. This phase was conducted using analog field equipment. These measurements indicate a layered, resistive section which can be correlated, in part, with the stratigraphic section known in the area from the Hadasim 1, Tzur Moshe 1, and Gaash 1 wells. It also gives evidence to the presence of the Mesozoic hinge line between sites 1 and 2.

It is recommended to extend the magneto-telluric survey as planned in the area with improved digital recording equipment.

1. INTRODUCTION

At the request of Sonol Israel Ltd., four magneto-telluric measurements were carried out by the Institute in the period 12-17 May 1974.

The measurements were located south of Netanya along a line directed approximately west-east from the sea coast to about 2 km. east of Tzur Moshe 1 well (Fig. 1).

These four measurements were conducted as a first phase of a survey planned for the Gaash-Caesarea area. They were carried out with analog recording equipment to try to obtain good evidence of the Mesozoic hinge line in the Netanya area. The analog equipment was used in order to obtain fast results, since the digital recorder will be supplied in late September.

2. THE MAGNETO-TELLURIC METHOD

The magneto-telluric method is based on the measurement of the variations of the magnetic and natural electric (telluric) fields at the surface of the earth. These variations are measured in time over a large band of frequencies (0.01-25 c/s).

The intensities of the magnetic and telluric fields vary with the frequency and reach very low values. These low values can be further affected by superimposed noise due to electric lines or industries which induce electric current in the ground. In some cases this noise is so strong that it is not possible to obtain the measurements of the natural fields.

3. MAGNETO-TELLURIC DATA

The four measurement sites were located roughly 3.5 to 1.5 km. apart south of Netanya (Fig. 1).

At each site the two orthogonal components of the telluric field (electric natural field) and the corresponding three components of the magnetic field (two in the orthogonal horizontal directions and one in the vertical direction) were recorded. The following field frequency bands were measured:

Band I ..... 0.01 - 0.125 c/s
Band II ..... 0.1 - 2.0 c/s
Band III ..... 1 - 25 c/s

The data were recorded on a system provided with an analog tape recorder.

The quality of the raw data is poor with a reduced number of magneto-telluric events and with a high level of noise superimposed on all the frequency bands recorded. This noise must be due to the electrical currents induced in the earth by the industrial activity in the Netanya area. The data quality becomes more affected by the noise level from west to east and because of this, the raw data measured at site 4 was insufficient to permit a quantitative interpretation and allowed only a tentative interpretation to be made.

It should be mentioned that in the vicinity of this site a radio transmitter was located.

In general the high frequency (1 - 25 c/s) band data was completely noisy and no valuable signal could be extracted in the processing for all the measured sites.

4. INTERPRETATION PROCEDURE

The field data recorded on analog tapes were processed at our computer center. In the first step the analog tapes were converted to digital data followed by a process of subsampling and anti-aliasing filtering. In the processing of the digital data a computer program AVESPEC was applied. This program based on tensorial computation, permits the calculation of the apparent resistivities at the surface of the earth for the principal directions of the structure (strike and dip) as function of the frequencies (periods).

The values of the apparent resistivities depend on the resistivities of the layers encountered in the earth by the electromagnetic waves. By increasing the measured period (T) an increased penetration is affected.

The apparent resistivities are plotted in bilogarithmic coordinates (apparent resistivity versus $\sqrt{T}$) for each measured site.

In order to obtain an earth model of resistivities and thicknesses, a method of quantitative interpretation based on curve matching is used, whereby changes on the apparent resistivity curve are compared to theoretical curves of different resistivity contrast. By using a multilayer case computer program, the theoretical curve for the computed model can be reconstructed.

For the quantitative interpretation the apparent resistivity curve corresponding to the case of the electric field parallel to the strike is considered.

5. RESULTS

The magneto-telluric results obtained at a measured site must be considered in the form that is essentially a greatly smoothed induction log showing the major conductive and resistive sections in the sedimentary column.

Two aspects of the interpretation of the magneto-telluric data one qualitative and the other quantitative, must be considered when discussing the results of the survey.

QUALITATIVE INTERPRETATION

By this interpretation the form of the apparent resistivities versus $\sqrt{T}$ curves (see appendix) obtained at the measured sites are considered. It is clear from this point of view that the curve obtained at site 1 differs greatly from that obtained at sites 2 and 3 which are similar. At site 4 only a part of the curve could be obtained due to poor data quality as explained before. The curves at sites 2, 3 and possibly 4 have the same general aspect and show that they are the expression of the same section of four resistivity layers. The curve of site 1 is completely different and is the expression of a five or six resistivity layer section.

Another qualitative aspect is shown on Figure 2 representing the apparent resistivity vs. T (T-period) cross-section, where the values of the calculated apparent resistivity obtained from the curves are plotted versus T for each site and isoresistivity curves are drawn. These curves can be drawn continuously in the area of sites 2 and 3 and 4. Between sites 1 and 2 no continuity of the curves could be followed in the lower part of the cross-section.

QUANTITATIVE INTERPRETATION

By this interpretation an earth model of resistivities and thicknesses was obtained for each site.

A magneto-telluric interpreted resistivity cross section (Fig. 3) was constructed by correlating the interpreted resistivities for each measured site. This cross section indicates between sites 2 and 4 a first, low resistivity (1.8-2.8 ohm-meters) layer followed by a thick layer of high resistivity (40(?)-70 ohm-meters). At depth this layer is followed by a very low resistive (0.4-1 ohm-meters) layer. The last layer detected by the magneto-telluric measurements does not have a well-defined resistivity and is probably higher than the values indicated on the section. For this general geologic section lower frequencies are needed to be able to better define the deepest layer. The first low resistivity layer corresponds to the Saqiye group with resistivities of 2 ohm-meters found in the Hadasim 1 and Tzur Moshe 1 wells. The corresponding calculated electric log of Hadasim 1 is integrated in the section. The detection of the superficial layer of 90 ohm-meters resistivity found in the well would need much higher frequencies to be recorded in the magneto-telluric measurements.

The limit separating the two first layers corresponds with the surface of the Judea group in the previously-mentioned wells. The resistivity found for the small section of the Judea group penetrated by Hadasim 1 well, is of the order of 180 ohm-meters with an indication of a lower value at the end of the log. This second magneto-telluric layer showing a resistive section, is considered to be indicative of hard rocks such as the limestone, dolomites, etc., of the Cretaceous and Jurassic rocks.

No further geologic correlations could be made for this part of the section.

It is possible that if the last layer has an effectively high resistivity, then it may represent the basement; but, these considerations are purely tentative due to the uncertainty in defining this layer from our measurements. This part of the section indicates that the deepest two surfaces become shallower to the west, a fact which cannot be explained with the known geological data. This interpretation must be verified with more measurements along new east-west oriented lines.

On site 1 the section indicates two possible interpretations of A and B.

Interpretation A indicates a first layer of 10 ohm-meters resistivity which corresponds in interpretation B to two layers of resistivity 7 and 35 ohm-meters. This section which must correspond to the Saqiye group (by correlation with seismic data and data from Gaash 1 well) indicates a higher resistivity than that found in the well. Of the two solutions, interpretation A fits the geologic data better. This layer (A) or layers (B) are followed by a layer of 3.3 or 2.6 ohm-meter resistivity.

In the Gaash well located about 8 km. to the south the Turonian-Cenomanian section found in the well has a resistivity of 4 to 9 ohm-meters, which is the same order as that found by magnetotellurics.

Such low resistivities under the Saqiye group may represent the Talme Yaffe and Gevar'am formations that are known to be composed of marlstone, chalk and shale.

For the rest of the resistivity layers defined by the magnetotellurics, correlation with geological data is difficult. The resistivity of the deepest layer is possibly high, but it could not be defined due to the same cause mentioned before. Speculatively this layer may represent the basement.

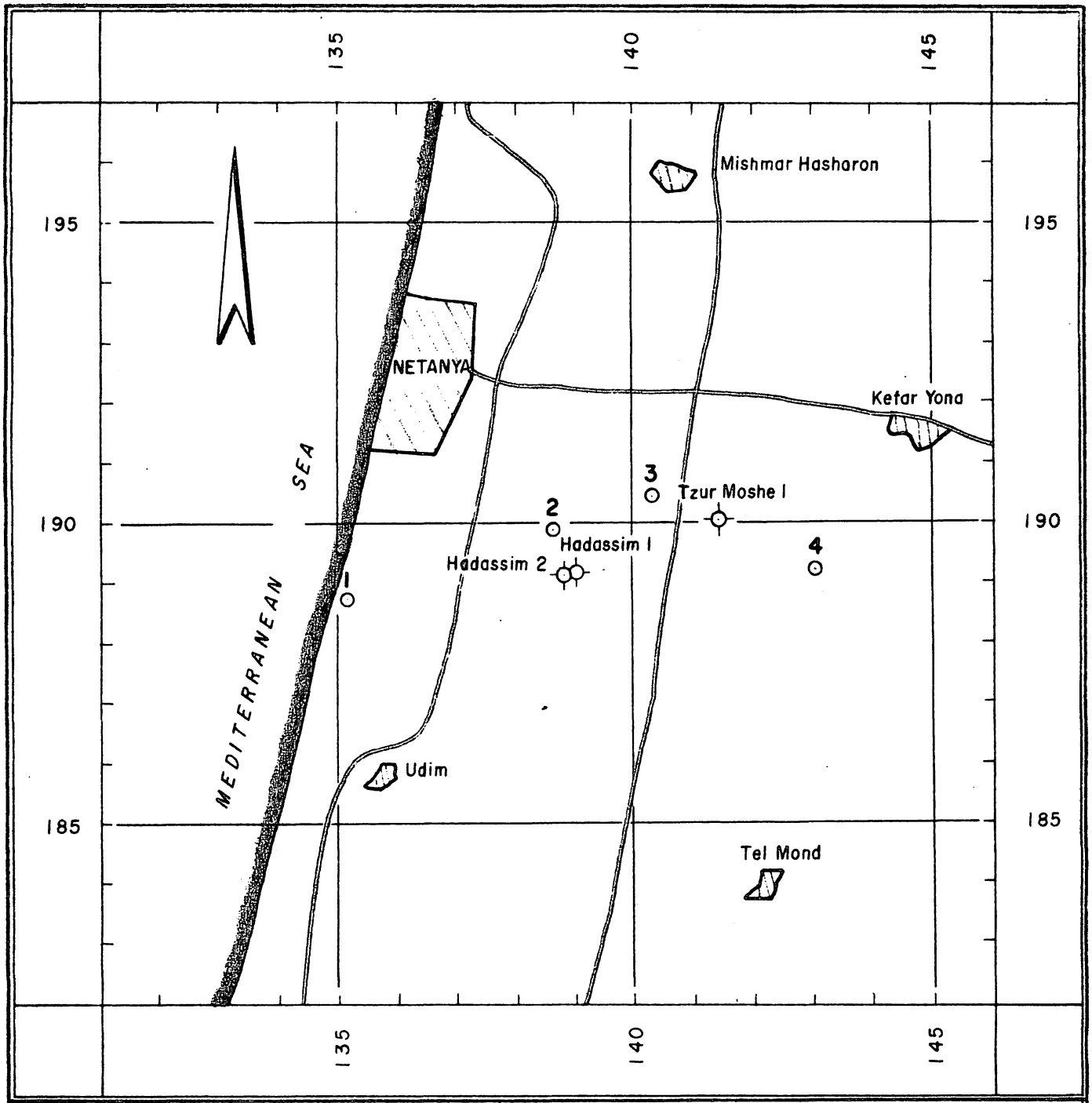
From this cross section it is clear that a discontinuity is present between measurement sites 1 and 2 and that different geologic formations are present on the two sides of the discontinuity evidenced by different resistivity ranges and thicknesses, determined by the magneto-telluric measurements.

It should be mentioned that because of the poor quality of data the quantitative interpretation can be affected by errors of about 15-20% in the determination of resistivities and depths. Another kind of error can be produced by the fact that determining a resistivity section from an apparent resistivity curve does not yield a unique solution. It is believed that the solution obtained is the best possible one, and takes into consideration the resistivity and geologic data known from previous work.

6. CONCLUSIONS AND RECOMMENDATIONS

The four magneto-telluric measurements conducted in the Netanya area gave some indication of the existence of the Mesozoic hinge line between sites 1 and 2. However, this reduced number of measurements are not sufficient for an accurate determination of this line.

It is recommended to continue with the complete survey as planned, using improved equipment with digital recording to try to upgrade the recorded data and to obtain lower frequencies for a better definition of the basement.



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Scale 1:100 000

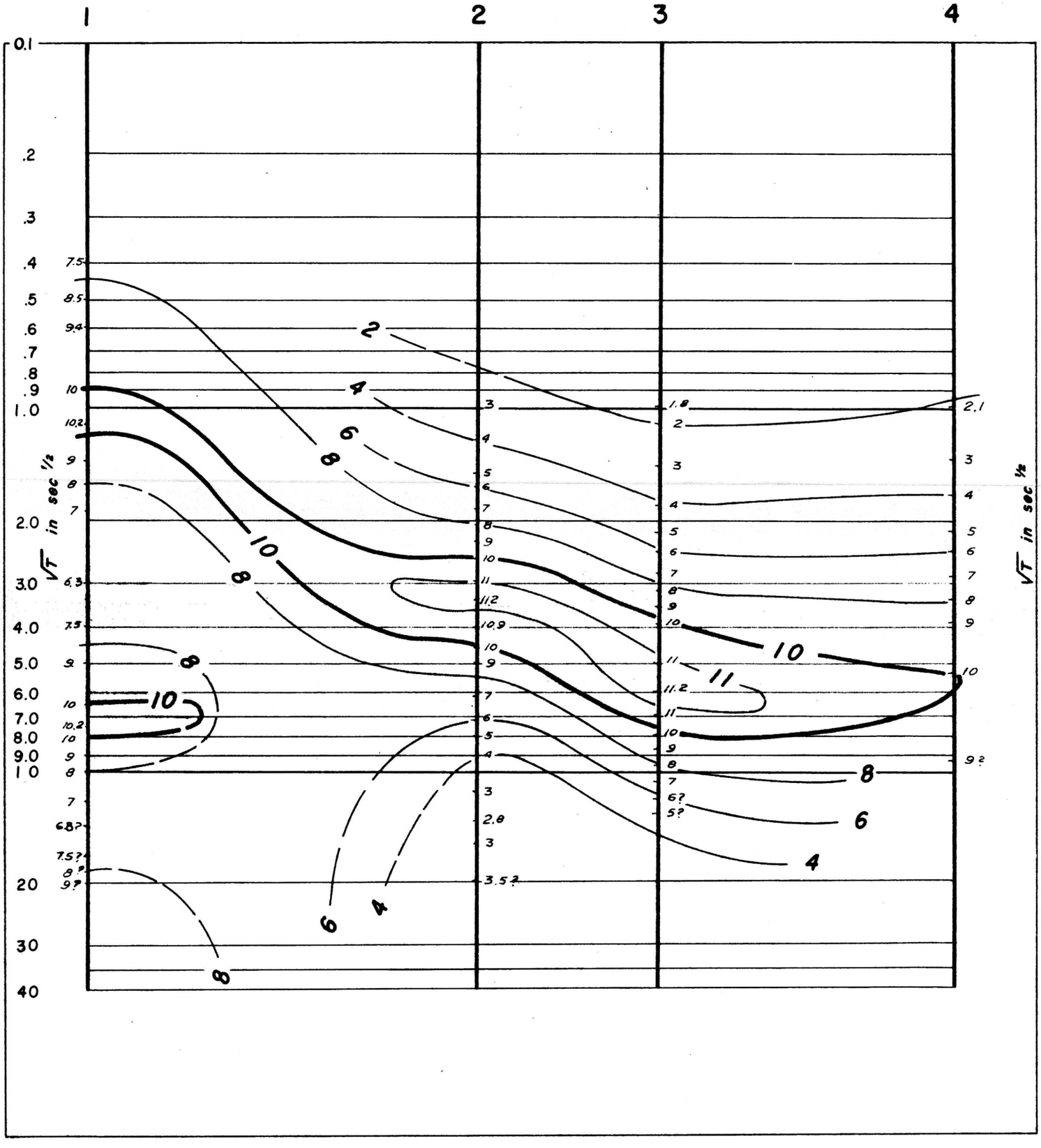
Magneto - telluric site 2

Well

AUGUST 1974

MT/118/74 (52057)

Figure 1



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APPARENT RESISTIVITY VS $\sqrt{T}$ CROSS SECTION

ELECTRIC FIELD PARALLEL TO STRIKE

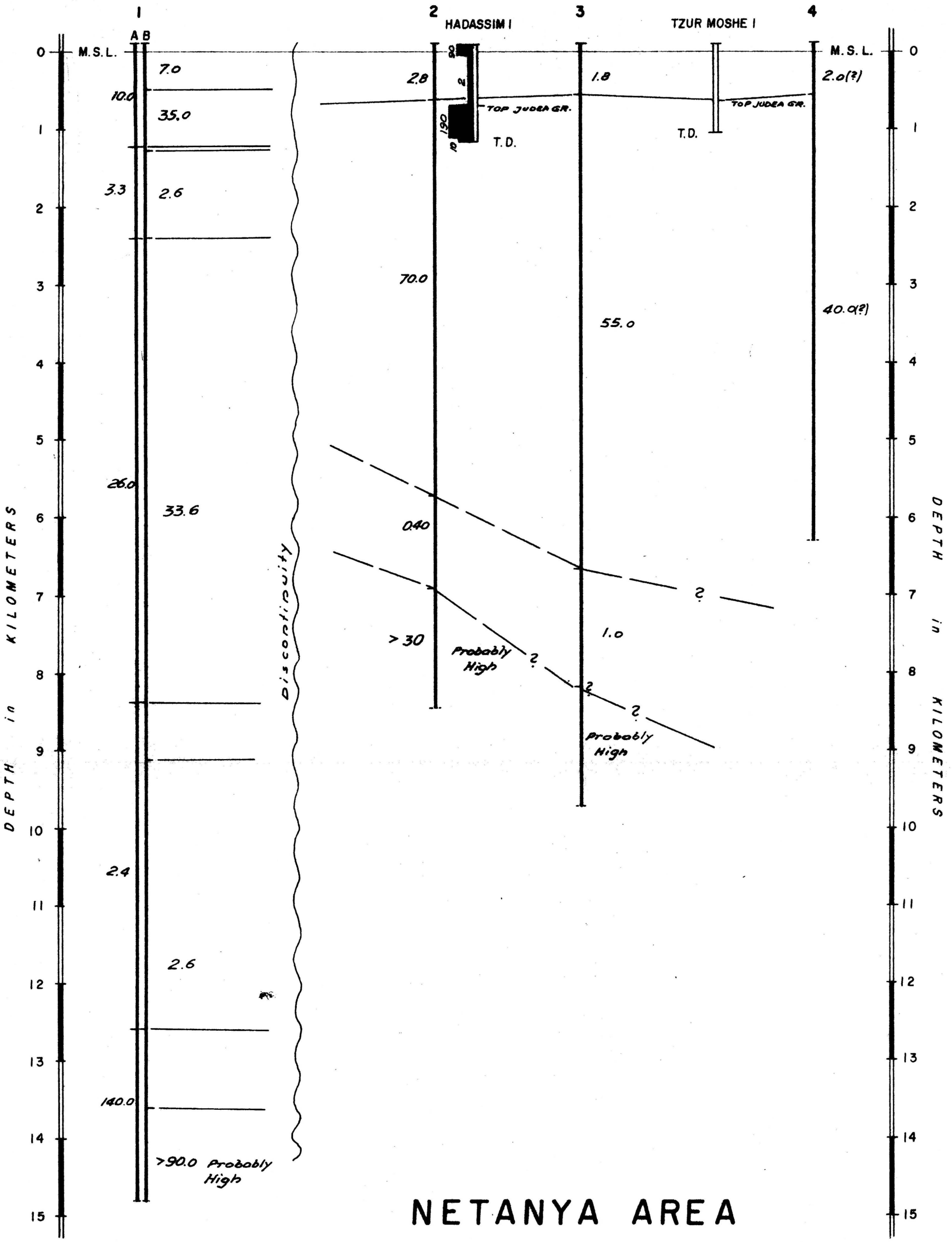
Horizontal Scale 1:50000

APPARENT RESISTIVITES IN OHM – METERS

AUGUST 1974

GI / 118 / 74 (52059)

Figure 2



NETANYA AREA

MAGNETO-TELLURIC INTERPRETED RESISTIVITY CROSS SECTION

HORIZONTAL SCALE 1 : 50 000
VERTICAL SCALE 1 : 50 000
RESISTIVITIES in OHM-METERS

Interpreted: A. KLANG
Date: August 1974

MT/118/74 (52 058)

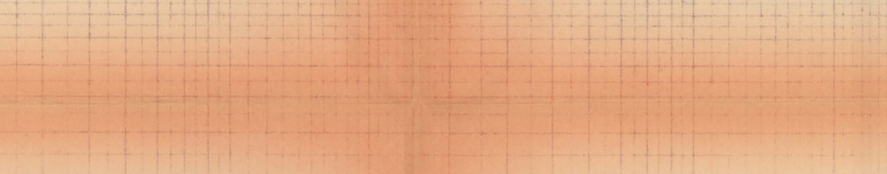
Figure 3

APPENDIX

Magneto-Telluric Apparent Resistivity Curves

M.T.P. Nos. 1 - 4

APPARENT RESISTIVITY Ω m



$\sqrt{T}$ in sec $^{1/2}$

CALC. RES Ω m							
10	3.3	26.0	2.4	140 PROBABLY HIGH	MODEL A
7	35	2.6	33.6	2.6	>90.0 PROBABLY HIGH	MODEL B
480	1200	2300	9100	13600	m
	1200	2400	8400	12600	m

M.T.P. No. 1

FILE No.

JOB No. MT/118/74 (52060)

LOCATION: SOUTH NETANYA

COORDINATES: 135 200 — 188 700

ELEVATION: 40 m

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APPARENT RESISTIVITY Ω m

$\sqrt{T}$ in $sec^{1/2}$

CALC RES Ω m

28	70	0.40	> 30 PROBABLY HIGH
630		5700	6900

m

M.T.P. No. 2

FILE No.

JOB No. MT/118/74 (52061)

LOCATION: SOUTH NETANYA

COORDINATES: 138700 — 189950

ELEVATION: 30 m.

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APPARENT RESISTIVITY Ω m

$\sqrt{T}$ in sec $^{1/2}$

CALC RES. Ω m	1.8	55.0	1.0	PROBABLY HIGH
	550	6650	8200 ?	m.

M.T.P. No. 3

FILE No.

JOB No. MT/118/74 (52062)

LOCATION: SOUTH NETANYA

COORDINATES: 140 400 — 190 400

ELEVATION: 40 m.

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APPARENT RESISTIVITY Ω.m.

$\sqrt{T}$ in $sec^{1/2}$

CALC RES Ω.m

2

40

560

m.

M.T.P. No. 4

FILE No.

JOB No. MT/118/74 (52063)

LOCATION: SOUTH NETANYA

COORDINATES: 143 050 — 189 150

ELEVATION: 50 m

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